

Canonical quarks, physical currents and strong CP

A representation-enforced Nelson–Barr assignment with a shared adjoint and fixed mixing frames

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Tree determinant protection and exact canonical matching at a specified flavor vacuum. Higher operators, quantum strong CP, observed flavor fits and golden protection remain separate requirements.

The canonical result

$$\bar{\theta}_{\text{tree}} = 0, \quad J = 1.13669 \times 10^{-6}$$

The same shared-adjoint scalar vacuum now admits a different, explicitly specified fermion assignment with physical weak CP and an exactly vanishing strong phase at tree level. The product-family representations enforce a universal bare Higgs contraction and forbid every Higgs-to-heavy Yukawa vertex at renormalizable order. All ten allowed renormalizable mass and Yukawa coefficients in each electric-charge sector are present and nonzero. The resulting Nelson-Barr mass structure is derived from this assignment, rather than imposed by setting an allowed renormalizable vertex to zero. The adjoint still connects the heavy source labels in both sectors. This changes the fermion representations of the previous construction; it is not a claim that canonical normalization repairs the previous mass matrices.

At the retained scalar background the new canonical benchmark has six nonzero light masses and a decoupling weak invariant $J = 1.13669458335\text{e-}6$. The CP partner has the opposite J and the same masses and mixing magnitudes. Neither branch uses a CKM target, a golden coefficient, an observed mass or a phase as an optimization input. The benchmark mixing magnitudes are $|V_{us}| = 0.0458605886$, $|V_{cb}| = 0.0260342476$ and $|V_{ub}| = 0.0020441921$. They do not constitute an observed CKM fit. The reference light Yukawa spectra are almost degenerate; a separate common-messenger-scale deformation demonstrates a canonical hierarchy mechanism, without claiming that its scales are dynamically selected or that its ratios reproduce observed quark masses.

The result has three precise parts: a representation-enforced renormalizable determinant protection, a canonical formula separating mass scales from mixing frames, and exact finite-scale correlations among charged currents, neutral currents and Higgs couplings. A constructive fixed-mass test also shows what the protection does not accomplish: positive allowed kinetic terms preserve the strong determinant phase while moving all four weak mixing directions. Thus protecting strong CP and predicting the golden relation are distinct mathematical obligations.

Why the previous assignment needed a determinant audit

$$\det D(v) = v^3 \det M \det Y_0, \quad \bar{\theta}_{\text{old}} = -2.46264$$

In the previous completion, the twelve-by-twelve Dirac matrix has a Higgs row connecting the ordinary doublets to heavy singlets, followed by nine heavy mass rows. In the right-handed basis consisting of three bare quarks and nine heavy quarks, the Higgs row is $T = [0, h, 0, 0]$ and the heavy row is $F = [C, M]$. Put $N = M^{-1} C$. The exact zero-Higgs null frame is $E = [I; -N](I + N^\dagger N)^{-1/2}$. The dimension-four light Yukawa is $Y = T E = Y_0(I + N^\dagger N)^{-1/2}$, with $Y_0 = -h N_u$. This is exact in the source-to-heavy ratios and is a decoupling statement in the Higgs-to-heavy ratio.

Block elimination gives $\det D(v) = v^3 \det M \det Y_0$. The phase of this determinant is independent of the Higgs scale. More generally any positive Hermitian canonical metrics multiply D from the left and right by inverse square roots with positive real determinants. They cannot change $\arg \det D$. A chiral phase rotation can make mass eigenvalues positive but shifts the microscopic QCD theta angle through the anomaly; it does not remove the invariant strong phase. With $\theta_{\text{QCD}} = 0$ in the original real-coupling basis, the previous assignment gives $\bar{\theta} = -2.46263764551$ radians. Its CP partner has the opposite phase. This is a tree-level result; no extra QCD mechanism was part of that assignment.

For an independent source-level check the nine-by-nine determinants reduce to six-by-six blocks. The up block is $[[A, 0.71 L], [-0.2 B S\text{-dagger}, 1.3 I - 0.04 B B\text{-dagger}]]$. The down block is $[[-0.2 A S, 0.71 L], [B, 1.3 I]]$. The determinant coefficient in each sector is minus h^3 times 2^3 times the determinant of its block. This removes the first heavy mass and its reverse link coefficient from the determinant audit. The reduced coefficients agree with the direct mass determinants to relative error below $3e-12$. Heavy decoupling and positive wavefunction normalization cannot suppress this phase.

The new representation assignment

Retain the declared scalar group $3_{A6_u} \times 3_{A6_d} \times 3_{A6_H} \times C6_A \times C6_B$ and CP, and the scalar fields $L = (3_u, 3_{\bar{d}})$, $A = (3_u, 3_{\bar{H}})$, $B = (3_d, 3_{\bar{H}})$ and $S = 8_H$. Their phase charges are respectively (0,0), (1,0), (0,1) and (-1,1). The ordinary left doublets and both bare right-handed quark triplets now transform as neutral 3_u . In each charge sector introduce three vectorlike electroweak-singlet triplets $D = (3_d; 0,0)$, $H_A = (3_H; -1,0)$ and $H_B = (3_H; 0,-1)$. These are color triplets with charge $2/3$ in the up sector or $-1/3$ in the down sector. The Higgs is a flavor and phase singlet. Its conjugate occurs in the up Yukawa in the usual way.

The constant kinetic endomorphism of each irreducible triplet is a scalar by Schur's lemma. The distinct heavy family representations or phase charges forbid constant cross kinetic terms. Their positive constants can be absorbed into field normalization and the real mass and Yukawa coefficients. Canonical renormalizable bare metrics therefore define a justified convention for this field content. This does not exclude higher-degree kinetic covariants after flavor breaking.

The full list of fermion contractions is short enough to audit exhaustively. There are three real diagonal vectorlike masses. The bare right-handed triplet couples to D through L-dagger and to H_A through A-dagger. D and H_B couple through B and B-dagger with independent real coefficients. H_A and H_B couple through S and S-dagger with independent real coefficients. Finally the ordinary doublet and bare right-handed triplet have the unique Higgs contraction $y I$. There is no renormalizable source contraction from the bare right-handed 3_u to H_B . There is no bare Higgs contraction from the left-handed 3_u to any heavy right-handed representation. Both observations follow from the simultaneous family labels and phase charges.

The operator receipt enumerates all nine mass/source contractions using charges modulo six and certifies their multiplicities with the exact 1,080-element group. The exact triplet endomorphism multiplicity is one; the adjoint has no singlet; the triplet-adjoint-triplet singlet multiplicity is one. Thus the ten real coefficients do not hide arbitrary component matrices. Their overall values remain independent physical inputs. The scalar potential and its retained 70-real-field local minimum are unchanged. The quantum corrections from the changed fermions, global vacuum selection and Higgs-portal-induced vacuum shifts are not calculated here.

A determinant protected by the interaction graph

$$\det D(v) = (vy)^3 \det M, \quad WMW^{-1} = (WMW^{-1})^\dagger$$

In the heavy order D, H_A, H_B , write $M = [[m_D I, 0, b B], [0, m_A I, s S], [c B\text{-dagger}, t S\text{-dagger}, m_B I]]$ and $C = [\lambda_L L\text{-dagger}; \lambda_A A\text{-dagger}; 0]$. The exact full Dirac matrix is $D(v) = [[v y I, 0], [C, M]]$. Every displayed coefficient is present. The absence of its upper-right Higgs block is enforced at renormalizable order. Consequently $\det D(v) = (v y)^3 \det M$ and is independent of the complex bare-heavy source column C.

The heavy interaction graph is a tree with two edges. For b, c, s, t positive, the constant diagonal matrix W with triplet weights $\sqrt{c/b}$, $\sqrt{t/s}$, 1 makes $W M W^{-1}$ Hermitian. Its two off-diagonal blocks are $\sqrt{bc}$ B and $\sqrt{st}$ S and their Hermitian conjugates. Therefore $\det M$ is real exactly, for every scalar background. On the connected branch where this Hermitian matrix is positive definite, its determinant is positive. With positive y_u and y_d , v positive and microscopic $\theta_{\text{QCD}} = 0$, the product of the up and down full determinants is positive and $\bar{\theta}$ vanishes exactly at tree level. The positivity domain is a stated branch condition; a determinant crossing through zero can change its sign and is not covered by a blanket claim about all backgrounds.

The benchmark's Hermitian-similar heavy matrices lie inside this positive region. A separate 80-digit determinant check reconstructs the real coefficients as exact decimal rationals before multiplication; its imaginary residual is below $1e-60$. An independent 80-digit full-spectrum check uses the saved floating matrices and retains their minute rounding asymmetry explicitly. Positive Hermitian changes of any bare fermion metrics preserve the determinant phase because their canonical normalization determinants are positive. This determinant protection is stronger than the canonical-metric convention and does not require the weak mixing matrices to remain fixed.

This is a realization of the established Nelson–Barr mechanism. The contribution of this calculation is the explicit product-family assignment, complete contraction census and shared-adjoint matching at our retained vacuum. General Nelson–Barr determinant protection, singlet-induced nonunitary weak mixing and flavor-changing neutral currents are established physics, not claimed as new discoveries. Primary context is Cherchiglia and Nishi, Consequences of vector-like quarks of Nelson-Barr type, arXiv:2004.11318, and Morozumi, Shimizu, Takahashi and Umeeda, Effective theory analysis for vector-like quark model, arXiv:1801.05268. Their URLs are retained in the scientific receipt.

Canonical mass scales and fixed mixing frames

$$y_i(\eta) = \frac{y\eta}{\sqrt{\eta^2 + \sigma_i^2}}, \quad P_i(\eta) = P_i(1)$$

The exact zero-Higgs null metric is $K = I + N^\dagger N$, where $N = M^{-1} C$, and the light Yukawa is $Y = y K^{-1/2}$. It is positive Hermitian in the common ordinary 3_u frame. Complex noncommuting up and down metrics produce physical weak CP even though each determinant is positive. Their left spectral projectors give $|V_{ij}|^2 = \text{Tr}(P_i^u P_j^d)$ in the decoupling limit. This gives a physical frame directly from the interaction and its canonical metric, with no free unitary matrix inserted in the readout.

Multiply the entire heavy M of one sector by a positive η while keeping C and y fixed. If σ_i are the singular values of $M_0^{-1} C$, the light Yukawa magnitudes are exactly $y\eta / \sqrt{\eta^2 + \sigma_i^2}$. The spectral projectors of K and Y remain unchanged because $K = I + \eta^{-2} N_0^\dagger N_0$ is a scalar function of the same fixed Hermitian matrix. Independent positive scale changes in the up and down sectors therefore preserve their decoupling mixing magnitudes and weak J , provided the spectra remain nondegenerate. In the small- η regime the masses become inversely proportional to the σ_i . This proves a mass hierarchy mechanism and a separation of scales from frames for this universal Higgs assignment; it does not select η or predict the observed hierarchy.

The prescribed scan uses $\eta = 1, 0.1, 0.01, 0.001, 0.0001$, with no fit objective. It changes the six masses substantially while reproducing the same mixing magnitudes to below $2e-13$. The scalar vacuum is held fixed. The statement is a decoupling theorem: reducing heavy masses at a fixed

physical Higgs scale can invalidate decoupling, so finite-scale observables must instead be obtained from the full spectrum below.

Finite-scale pole normalization and physical currents

$$V = N_u^\dagger N_d, \quad Z_f = N_f^\dagger N_f, \quad h_{ij} = Z_{ij} m_j / v$$

For either fermion assignment define the full-rank heavy row F and Higgs row T , and let $G = F F^\dagger$. The exact left mass-squared matrix has blocks $[[v^2 T T^\dagger, v T F^\dagger], [v F T^\dagger, G]]$. At a light pole λ below the heavy gap, elimination of the singlet component gives $v^2 [T T^\dagger - T F^\dagger (G - \lambda I)^{-1} F T^\dagger] q = \lambda q$. The singlet component is $r = -v (G - \lambda I)^{-1} F T^\dagger q$. The normalized mode satisfies $q^\dagger [I + v^2 T F^\dagger (G - \lambda I)^{-2} F T^\dagger] q = 1$. Because this positive metric depends on the pole, one common exact three-by-three wavefunction metric cannot replace the full finite-scale calculation.

Expanding the pole equation yields the dimension-six left metric $I + v^2 Z_1$, with $Z_1 = T F^\dagger G^{-2} F T^\dagger$. Z_1 is positive semidefinite because it is a Gram matrix. The light Hamiltonian at this order is $v^2 K_{\text{left}}^{-1/2} H_0 K_{\text{left}}^{-1/2}$, with $H_0 = Y Y^\dagger$. The implementation obtains the full twelve-by-twelve singular spectrum directly and compares it with this derivative expansion. At $v = 0.03$ in the reference messenger units, the light-current magnitude approximation error is approximately $7.23e-9$ and the maximum relative light-mass error approximately $3.18e-9$. Across $v = 0.03$ and 0.1 the current error follows the expected fourth-order scaling. At very small v floating-point eigenframe errors set a numerical floor. No physical Higgs-to-messenger scale is fixed by this benchmark.

Let N_u and N_d be the ordinary-doublet rows of the three light left eigenvectors. The physical light charged current is $V_{\text{light}} = N_u^\dagger N_d$. The weak-isospin neutral currents are $Z_u = N_u^\dagger N_u$ and $Z_d = N_d^\dagger N_d$, after stripping the standard electroweak couplings and universal electric-charge contribution. Generally V_{light} is not unitary. Its row deficit has the exact positive decomposition $I - V_{\text{light}} V_{\text{light}}^\dagger = (I - N_u^\dagger N_u) + N_u^\dagger (I - N_d N_d^\dagger) N_u$. The column version follows by exchanging sectors. The off-diagonal neutral-current entries are consequently part of the same matching prediction, rather than optional additional matrices.

With all light and heavy states included, the charged-current matrix is a rank-three partial isometry and its two products are the full neutral-current projectors. The Higgs coupling in the full mass basis obeys the exact identity $h_{ij} = Z_{ij} m_j / v$. It follows from the top rows of $D R_j = m_j U_j$ and holds for all twelve modes in each sector. Tests check this relation, the neutral projectors, the positive deficit decompositions, the pole equation and the Higgs mass derivative independently. At finite scale the nine rephasing-invariant quartets need not have the same absolute value. A nonzero quartet certifies weak CP, while reporting one unitary J or extracting a unitary standard phase would discard part of the physical current information.

The precise protection boundary

$$\Delta\bar{\theta} = 3 \arg(y_d + c \det L / \Lambda^3)$$

The renormalizable assignment protects the full determinant phase, not the golden weak relation. On the ordinary right-handed 3_u space use $X = A^\dagger$ and $Z = B^\dagger L^\dagger$. The previous nine-covariant construction provides CP-compatible real polynomial kinetic generators at this vacuum. Twelve positive polynomial metric deformations preserve all six

decoupling masses while spanning four independent directions in $|V_{us}|$, $|V_{cb}|$, $|V_{ub}|$ and J . The smallest normalized response singular value is 0.5567; the maximum mass log residual is $1.34e-15$. Because these metrics have positive determinants, they leave the protected tree strong phase zero. A protecting mechanism for the golden relation must still constrain this kinetic operator space.

An allowed Yukawa operator provides an explicit limit to the determinant protection: a real coefficient multiplying $\det(L) \bar{Q}_L H d_R / \Lambda^3$ plus its Hermitian conjugate. The invariant $\det L$ is neutral under the declared family and phase symmetries, and the bare universal Higgs Yukawa is independently invariant. Their product is therefore allowed and CP even with a real coefficient. It is dimension seven. It shifts the universal down Yukawa from y_d to $y_d + c \det L / \Lambda^3$ and creates a strong phase $3 \arg(y_d + c \det L / \Lambda^3)$. No ordinary phase charge that retains both the bare Yukawa and the existing cubic $\det L$ scalar interaction can forbid this product: invariance of the cubic requires the phase of $\det L$ to be one.

At the retained background $\text{Im } \det L = 3.17850793318e-6$ in reference messenger units. For coefficient one and cutoffs 1,10,100 in those units, this single operator creates phases approximately $1.67e-5, 1.67e-8, 1.67e-11$ respectively. These are a scaling diagnostic, not a phenomenological cutoff determination. The leading expression is $3 c \text{Im } \det L / (y_d \Lambda^3)$. Quantum threshold contributions to strong CP and the complete higher-degree Yukawa operator space remain uncalculated. A vanishing tree determinant phase is not an all-orders solution of strong CP.

The fixed-background finite-Higgs spectra also do not establish the complete electroweak vacuum: Higgs portals allowed in the scalar action could shift the flavor condensates, and their coefficients were not part of the retained zero-Higgs scalar calculation. The determinant theorem holds throughout its positive branch regardless of those shifts, but the numerical mixing predictions would need to be recalculated. Global flavor-vacuum selection, the M22 origin of three ordinary doublets, observed masses and mixing, radiative stability and a protected golden relation remain separate requirements.

Reproduction

Run `OPENBLAS_NUM_THREADS=1 PYTHONPATH=python python python/develop_valentiner_canonical.py` from the repository root. It reads the disclosed previous scalar vacuum and produces `receipts/m22_interactions/valentiner_canonical.json`, including the exact operator census, old determinant obstruction, new determinant checks, finite-scale spectra and currents, mass-scale scan, opposite CP branch, fixed-mass kinetic countermetrics and quality operator. No scalar minimization or CKM fit is hidden in this driver. The canonical matcher is general for a full-rank n by $n+3$ heavy row with an invertible square heavy block and three ordinary doublets. The release includes the standalone paper, updated complete monograph, tests and complete tracked source archive.